

6. The geometry of matrices as functions

 [Bookmark this page](#)

Upgrade your course today

- ✓ You are taking the free version of this course.
- ✓ Upgrade today and, upon passing, receive your digital certificate signed by MIT faculty to highlight the knowledge and skills you've gained from this MITx course.

Upgrade

 Calculator

Lecture due Jan 24, 2024 20:30 IST

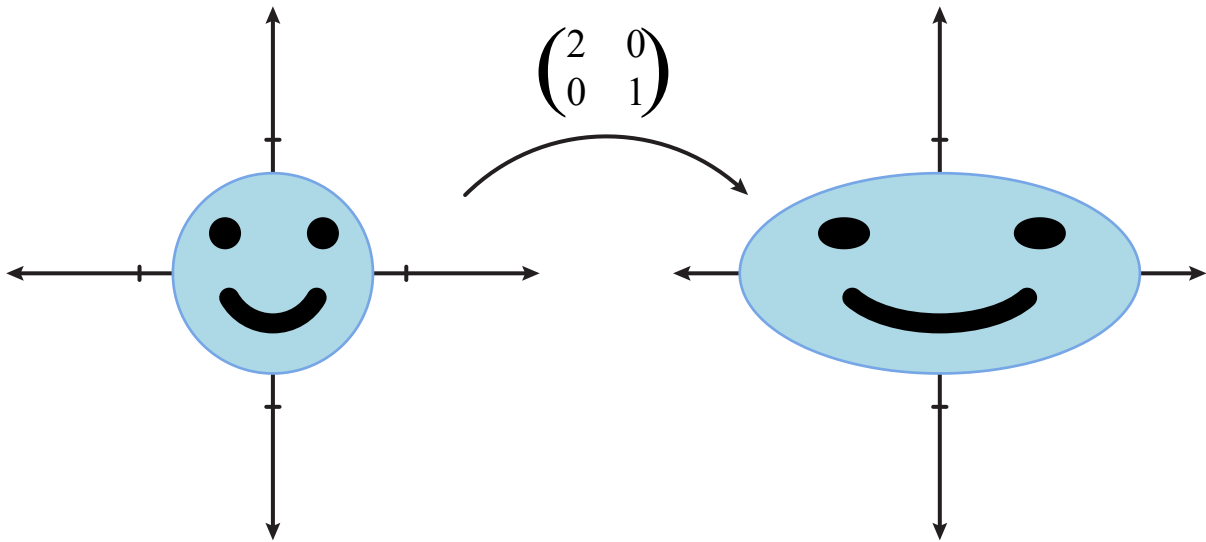
Imagine evaluating an $m \times n$ matrix $\mathbf{A}$ on every vector in the input space $\mathbb{R}^n$, to get vectors in the output space $\mathbb{R}^m$. To visualize it, draw a shape in the input space, apply $\mathbf{A}$ to every point in the shape, and plot the output points in the output space.

Problem 6.1 The matrix $\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$ represents a function $\mathbf{f}$ from $\mathbb{R}^2$ to $\mathbb{R}^2$. Determine what it does to the *standard basis* vectors $\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $\mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ of $\mathbb{R}^2$. Then depict what happens to a smiley face of unit area under this matrix function.

Solution: We have

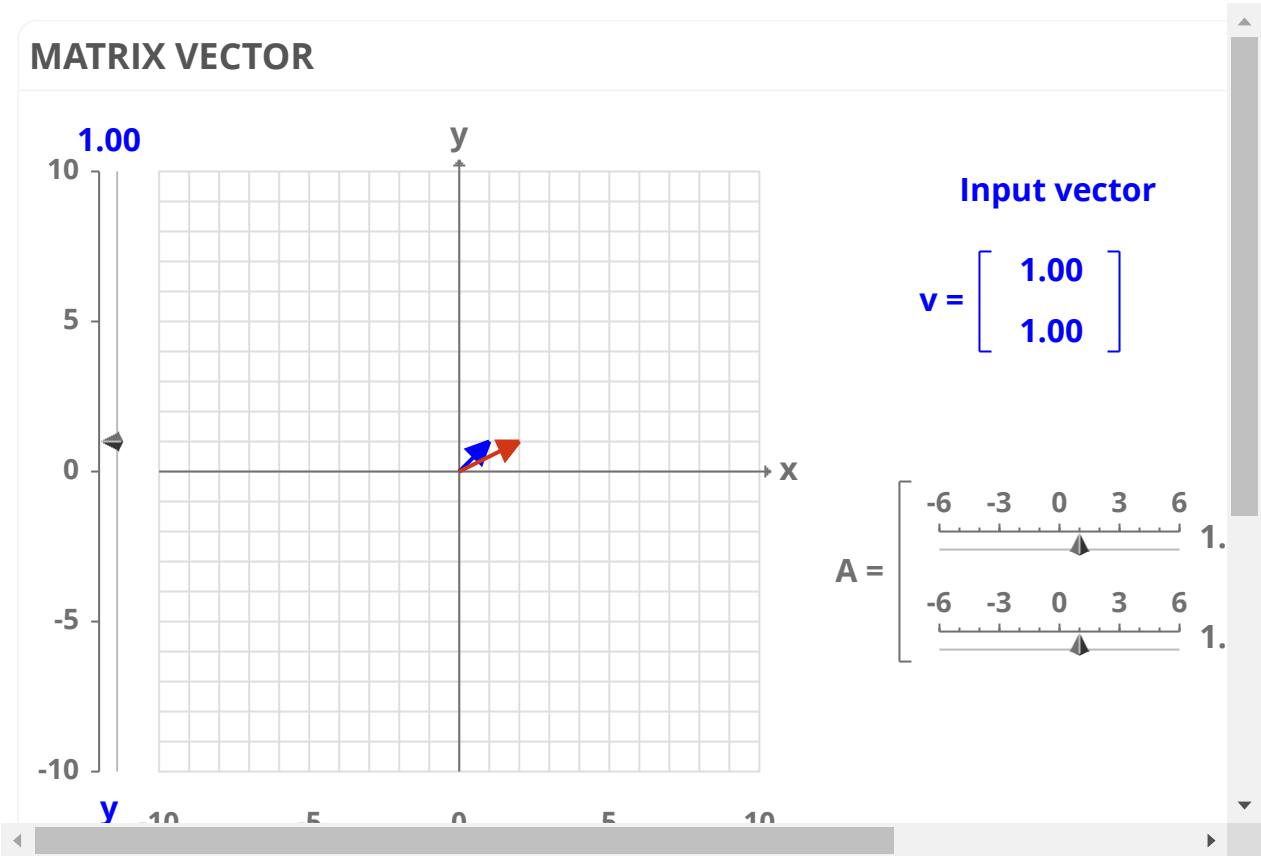
$$\mathbf{f}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \end{pmatrix}$$
$$\mathbf{f}\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

and the smiley with unit radius is stretched horizontally into a wide smiley of the same height.



Mathlet exploration

2/2 points (graded)



Use the mathlet above. Specify any arbitrary matrix $\mathbf{A}$ by changing the entries with the sliders. Then answer the following questions.

1. Where does the matrix $\mathbf{A}$ send the vector $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$?

(Hint: To create the input vector $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$, use the x and y sliders next to the vector graph output. Then try changing the entries of $\mathbf{A}$ and see what happens to the output vector.)

☒ To the vector that is in the first column of the matrix $\mathbf{A}$.

☐ To the vector that is in the first row of the matrix $\mathbf{A}$.

☐ To the vector that is in the second column of the matrix $\mathbf{A}$.

☐ To the vector that is in the second row of the matrix $\mathbf{A}$.



2. Where does the matrix $\mathbf{A}$ send the vector $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$?

(Hint: To create the input vector $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$, use the x and y sliders next to the vector graph output. Then try changing the entries of $\mathbf{A}$ and see what happens to the output vector.)

☐ To the vector that is in the first column of the matrix $\mathbf{A}$.

☐ To the vector that is in the first row of the matrix $\mathbf{A}$.

☒ To the vector that is in the second column of the matrix $\mathbf{A}$.

☐ To the vector that is in the second row of the matrix $\mathbf{A}$.



Solution:

The matrix $\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$ sends the vector $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ to the vector

$$\mathbf{A} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} a_{11} \\ a_{21} \end{pmatrix}$$

which is the first column of the matrix $\mathbf{A}$.

The matrix $\mathbf{A}$ sends the vector $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ to the vector

$$\mathbf{A} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} a_{12} \\ a_{22} \end{pmatrix}$$

which is the second column of the matrix $\mathbf{A}$.

1/14/24, 4:43 AM

1 Elimination and solving linear systems with matrices | Unit 1: Linear Algebra, Part 1 | Linear Algebra and NxN Systems | MITx Online

← Previous

Next →

Submit

You have used 1 of 3 attempts

ⓘ

Answers are displayed within the problem

© All Rights Reserved

© MITx Online. All rights reserved except where noted.

[About Us](#)

[Terms of Service](#)

[Privacy Policy](#)

[Honor Code](#)

[Accessibility](#)

There are no posts in this topic yet.

